Do regional calendars help electricity forecasting?

Controlled evaluation on Indian state supplied-energy data

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Research in progress

Why study this?

Moving dates

Festival dates vary across years, and regional observances can differ.

Forecasting challenge

Pretrained models may use calendar inputs differently, even on the same cases.

Precise target

Daily energy met/supplied in GWh: what the system delivered, rather than latent unconstrained demand.

Data and evaluation rules

- Three states: Karnataka, Delhi and Maharashtra.
- Fixed source snapshot through 28 October 2026; sources remain separate.
- 730-day context; individual lead days 1, 7 and 14.
- Chronological selection and purged labels; score recorded labels only.

Historical outcomes are exploratory; matched cases make comparisons meaningful.

Models and comparison design

Local models

Ridge and LightGBM

Rich causal history features

Annual versus monthly refits

Foundation models

Chronos-2 and TimesFM 3

Pinned pretrained weights

Historical context + known calendar

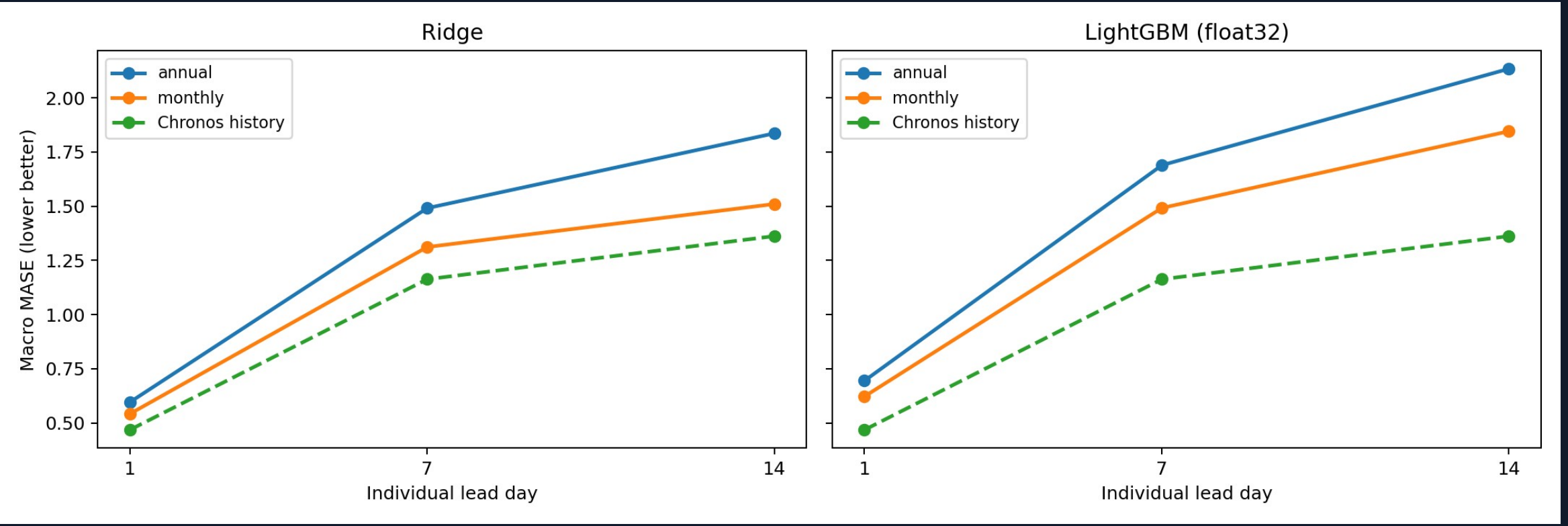
Fair comparisons

Same state/origin/lead cases

Pre-year parameter selection

Regional transfer tested separately

Finding 1: refitting matters



Monthly updates improve several aggregates, but can worsen individual festival-year results.

Partial 2026, identical cases. Ridge version 1; tree float32 version 2; cached Chronos history.

Controls before making a calendar claim

Hold dimensions equal

Generic vs regional dates

One scalar window or seven offset channels

Add negative controls

One/seven zero channels

Active weekday/season-matched placebo dates

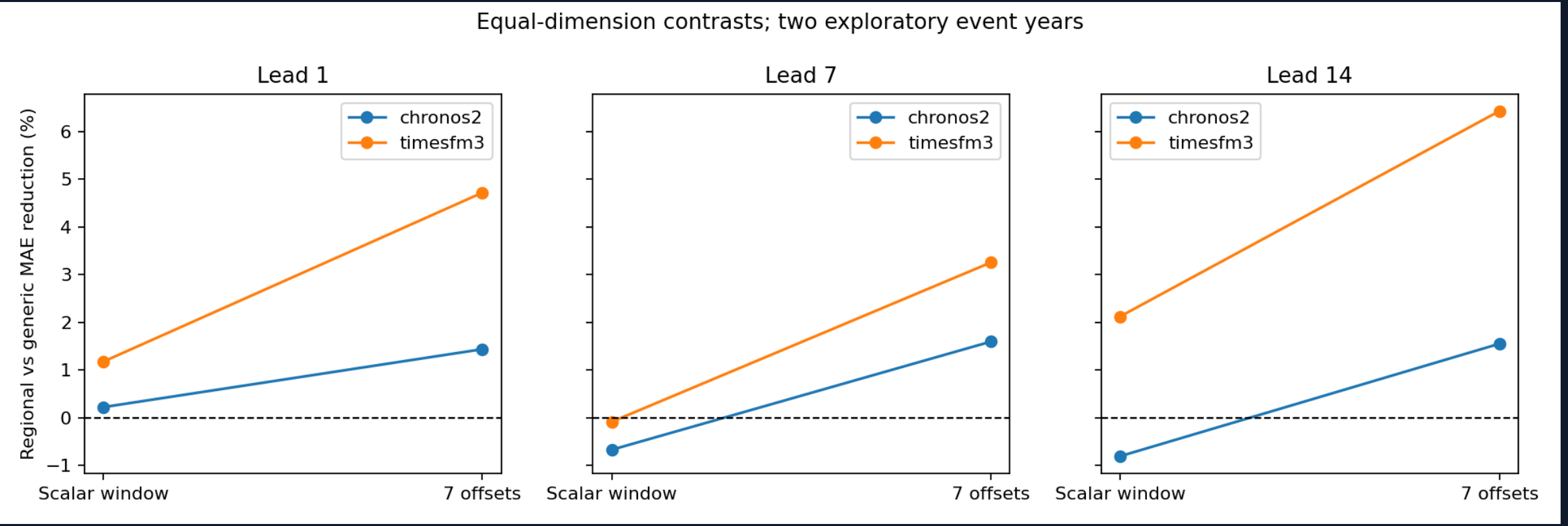
Keep evidence fixed

Same 150 historical cases per arm

Two models, nine arms

Individual years and failures retained

Finding 2: representation changes the answer



Regional gains depend on the model, lead and representation.

Two historical event years; equal channel counts within each comparison; positive = lower MAE.

Quality checks found issues worth reporting

Data revisions

Archived values differ from later snapshots by up to 61.5 GWh in the reviewed cases.

Reporting availability

Two tested issue-date snapshots lag by 2–3 calendar days; matched errors increase about 3.9–11.3%.

Numerical stability

Tiny rolling-arithmetic differences affect trees. A separate float32 contract reproduces the checked examples.

32 tests pass; separate-environment/software review is not external peer approval.

Prospective validation: frozen before outcomes

30 October

Forecasts frozen in Git

Observed context ends 28 Oct

31 Oct–11 Nov

Eligible target dates

Outcomes still arriving

12 November onward

Date-gated collection/scoring

No prediction changes

The full target period is not complete at this presentation.

What the study supports today

- A reproducible empirical comparison with meaningful controls and retained negative results.
- Model interfaces, adaptation policy and source vintages can change the interpretation.
- Pooling regions can help; withholding the target region generally hurts the tested regression models.

A bounded research study—not a claim of a new architecture or universal festival gain.

What happens next?

New outcomes

Score the immutable forecasts after 11 November, without retuning.

Human and source review

Permission email sent by the presenter; response pending. Independent reviewer still needed.

Submission decision

Review the evidence and limitations before deciding whether a journal submission is justified.

Sources and questions

- Robbie Andrew: Indian Energy and Emissions Data; upstream GRID-INDIA/POSOCO.
- Hunt & Bloomfield: original corrected state dataset, Zenodo 14983362 (CC BY 4.0).
- Chronos-2: arXiv 2510.15821; TimesFM: Google Research official repository/model.
- Kreusel et al.: calendar encoding study, arXiv 2610.13676.

All displayed charts are generated from this project’s recorded aggregate analyses.